DERIVATION OF MODELS

We provide an in-depth discussion of two market impact modeling approaches: the Almgren & Chriss path dependent approach and the I-Star cost allocation approach.

Almgren & Chriss—Market Impact Model

The Almgren & Chriss (AC) market impact model is path dependent based on the actual sequence of trades and executions. Cost is computed as the difference between the actual transaction value of the sequence of trades and the transaction value that would have occurred had all the trades been executed at the arrival price. The AC model follows closely to the graphical representative shown in the price trajectory graphs of Madhavan (2000).

The cost function corresponding to the AC model is:

$$Cost = Side \cdot \left(\sum x_i P_0 - \sum x_i p_i\right) \tag{4.1}$$

where,

$$Side = \begin{cases} +1 & Buy\ Order \\ -1 & Sell\ Order \end{cases}$$

$$x_i = shares\ traded\ in\ the\ i^{th}\ transaction$$

$$p_i = price\ of\ the\ i^{th}\ transaction$$

$$P_0 = arrival\ price$$

$$\sum x_i = total\ shares\ traded$$

It is important to note here that this calculation only incorporates the trading related transaction cost component and not potential opportunity cost. For purposes of building a market impact model, one of the basic underlying assumptions is that all shares of the order X will be transacted, e.g., $\sum x_i = X$.

The Almgren & Chriss model computes market impact cost for each individual trade. The entire sequence of trades is then rolled up to determine total value traded and total trading cost. Because this approach is based on the sequence of trades the model is referred to as a path dependent approach. Additionally, because total cost is derived from trade level data it is also often referred to as a bottom-up approach.